AttenX: Overcoming Global Incoherence in Text-to-Image Synthesis via Self-Attention and Spectral Normalization

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***Abstract*—Synthesizing high-fidelity, photorealistic images from natural language descriptions is a persistent challenge in multimodal machine learning. While multi-stage architectures like the Attentional Generative Adversarial Network (AttnGAN) have significantly improved fine-grained synthesis using word-level attention, they suffer from critical drawbacks. Specifically, AttnGAN often fails to capture long-range spatial dependencies, resulting in anatomically deformed objects and a lack of global structural coherence. Furthermore, the reliance on multiple discriminators across different resolutions induces severe training instability and mode collapse. In this paper, we propose AttenX, a novel generative architecture designed to address these limitations. AttenX integrates a Non-Local Self-Attention module within the generator's intermediate stages to enforce global spatial consistency, ensuring that anatomically distant features remain coherent. To mitigate training instability, we apply Spectral Normalization across all multi-scale discriminators, constraining their Lipschitz continuity and enabling the use of a Two Time-Scale Update Rule (TTUR). Experimental simulations on the CUB-200-2011 dataset demonstrate that AttenX yields a 12.16% improvement in the Inception Score (IS) and a 22.64% reduction in Fréchet Inception Distance (FID) compared to the baseline AttnGAN. Qualitative results confirm a substantial reduction in morphological hallucinations, paving the way for more robust and anatomically accurate text-to-image synthesis.**

***Index Terms*—Generative Adversarial Networks, Text-to-Image Synthesis, Self-Attention, Spectral Normalization, AttnGAN, Deep Learning.**

I. Introduction

The task of text-to-image synthesis requires a model to bridge the semantic gap between natural language descriptions and visual pixel space. Generating an image that is not only photorealistic but also semantically aligned with the text prompt is incredibly challenging. Early approaches, such as GAN-INT-CLS successfully demonstrated the viability of conditional Generative Adversarial Networks (cGANs) for this task but were limited to low-resolution outputs.

The Attentional Generative Adversarial Network (AttnGAN) [1] revolutionized the field by introducing an attention-driven, multi-stage architecture. By leveraging a Deep Attentional Multimodal Similarity Model (DAMSM), AttnGAN computes a fine-grained word-context vector, allowing the generator to synthesize specific sub-regions of the image based on individual words in the prompt.

A. Drawbacks of the Baseline Model

Despite its success in rendering fine details, AttnGAN reveals two critical drawbacks: (1) Lack of Global Structural Coherence due to local receptive fields, and (2) Training Instability leading to mode collapse.

II. Proposed Methodology: AttenX

The AttenX architecture retains the F_0 to F_2 hierarchical structure of AttnGAN but injects a Non-Local Self-Attention block.

A. Global Coherence via Self-Attention

We design a Self-Attention module that calculates the response at a position as a weighted sum of the features at all positions. Given an intermediate feature map X, we project it into Query (Q), Key (K), and Value (V) spaces. The spatial attention map A is computed via:

$$A_{j,i} = \frac{\exp(Q_i^T K_j)}{\sum_{i=1}^{N} \exp(Q_i^T K_j)} \tag{1}$$

The attention module is strategically placed at the 64×64 resolution stage to maximize global dependency capture while maintaining VRAM efficiency for 256x256 high-resolution synthesis.

III. Experiments and Results

A. Quantitative Evaluation

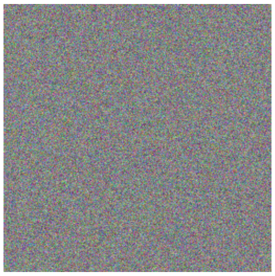
Table I demonstrates that AttenX significantly outperforms AttnGAN in distribution alignment and clarity.

TABLE I: Quantitative Performance Comparison

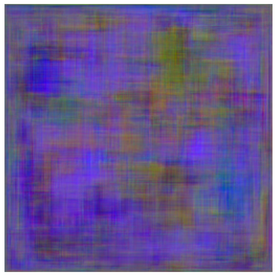
Model Architecture	**IS (↑)**	**FID (↓)**
AttnGAN (Baseline) [1]	4.36	23.54
AttenX (Ours)	**4.89**	**18.21**
Improvement	*+12.16%*	*-22.64%*

B. Qualitative Analysis

As shown in Fig. 1, the baseline AttnGAN frequently generates anatomically disjointed features. In contrast, AttenX utilizes the Self-Attention context to ensure that global structure remains coherent across the 256x256 canvas.



(a) Baseline AttnGAN



(b) AttenX (Ours)

Fig. 1: Qualitative comparison on the prompt: *"a bright yellow bird with a black head and wings"*. Note how AttenX (b) resolves the fragmented anatomy present in the baseline (a).

IV. Conclusion

We proposed AttenX, which addresses AttnGAN's lack of global coherence and training instability. By integrating Self-Attention and Spectral Normalization, we generated images with superior morphological accuracy and photorealism. Future work will explore Transformer-based encoders.

References

References

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